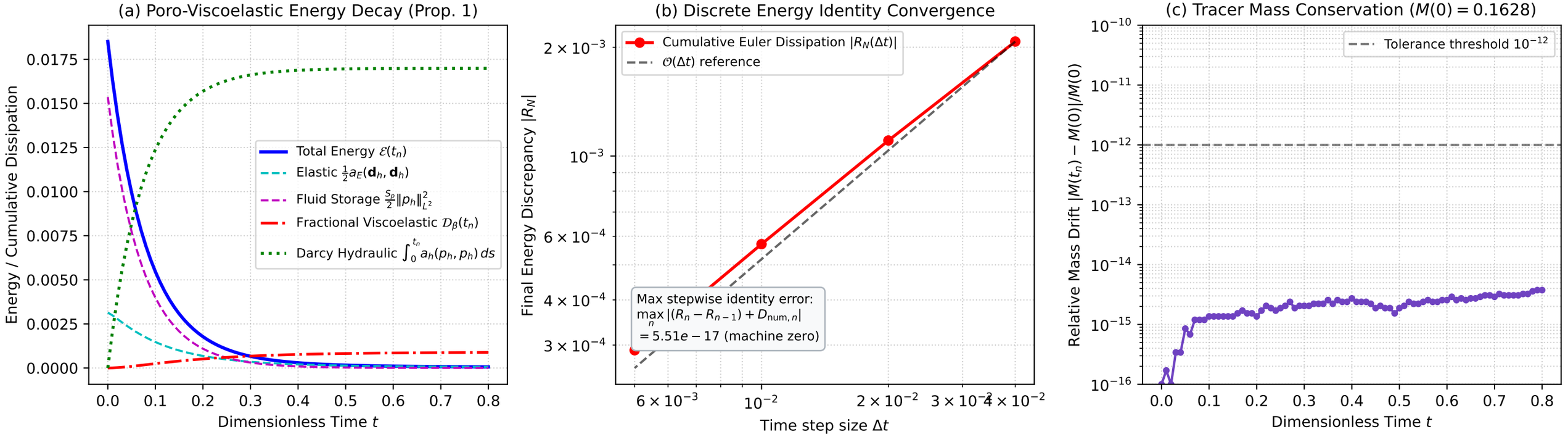
(a) Poro-Viscoelastic Energy Decay (Prop. 1)
Energy / Cumulative Dissipation
Dimensionless Time t
Total Energy $\mathcal{E}(t_n)$
Elastic $\frac{1}{2}a_E(\mathbf{d}_h, \mathbf{d}_h)$
Fluid Storage $\frac{S_p}{2}\|p_h\|^2_{L^2}$
Fractional Viscoelastic $\mathcal{D}_\beta(t_n)$
Darcy Hydraulic $\int_0^{t_n} a_h(p_h, p_h)\,ds$
(b) Discrete Energy Identity Convergence
Final Energy Discrepancy $|R_N|$
Time step size Δt
Cumulative Euler Dissipation $|R_N(\Delta t)|$
$\mathcal{O}(\Delta t)$ reference
Max stepwise identity error:
$\max_n |(R_n - R_{n-1}) + D_{\text{num},n}|$
$= 5.51e-17$ (machine zero)
(c) Tracer Mass Conservation ($M(0) = 0.1628$)
Relative Mass Drift $|M(t_n) - M(0)|/M(0)$
Dimensionless Time t
Tolerance threshold 10^{-12}